

Calculus Skeleton Notes

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Average Rates of Change

Goals: Compute average rates of change, slopes of secant lines, and motivate the idea of limits

Before Class Videos: None today, starting next class

Example 1: I left my house at time $t = 0$. 30 minutes later, I was at UVic which is 12 km away.

a) What was my average speed?

b) Sketch a possible graph

c) How could I approximate my speed at $t = 10$ minutes?

Example 2: What is average rate of change of $f(x) = x^2$ on $[2, 4]$?

Example 3: Find the equation of the line through (x_1, y_1) and (x_2, y_2) .

Example 4: Consider the graph of a function $f(x)$. How do I write an equation for the secant line through $(x_1, f(x_1))$ and $(x_2, f(x_2))$?

Big Idea: As x_2 gets closer to x_1 the slope of the secant line approaches the slope of the tangent line at x_1 . We need to investigate and define this limiting process more precisely. This DESMOS visualizes the idea: <https://www.desmos.com/calculator/oi7poyhodr>

Suggested after class problems from APEX: None today, but review the Course Outline on Brightspace

1.1 - Limits

Goals: Describe limits graphically, numerically, and algebraically.

Before Class Videos: 1.1.1 and 1.1.2

Example 1: Consider these three functions:

$$f(x) = x + 1, \quad g(x) = \frac{x^2 - 1}{x - 1} = \frac{(x + 1)(x - 1)}{x - 1}, \quad h(x) = \begin{cases} x + 1, & x \neq 1, \\ 3, & x = 1. \end{cases}$$

How are these similar or different?

Definition:

$\lim_{x \rightarrow a} f(x) = L$ if $f(x)$ can be made arbitrarily close to L for all values of x sufficiently close to a .

Example 2: Compute the limit as $x \rightarrow 1$ of $f(x)$, $g(x)$, and $h(x)$.

Example 3: Use a table of values to estimate $\lim_{x \rightarrow 1} (x + 1)$.

Example 4: Use a table of values to estimate $\lim_{x \rightarrow 0} \frac{1}{x}$.

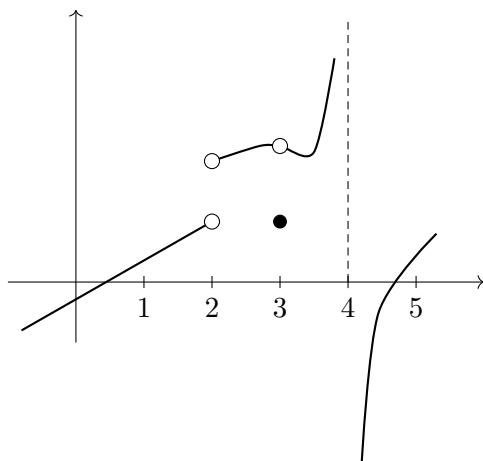
Example 5: Investigate

$$f(x) = \begin{cases} x, & x < 1, \\ x + 1, & x \geq 1. \end{cases}$$

Definition:

$\lim_{x \rightarrow a^+} f(x)$ means $f(x)$ gets arbitrarily close to L for all x sufficiently close to a from the right,
i.e. $x > a$.

Example 6: Use the graph to evaluate the indicated limits.



$$\lim_{x \rightarrow 2^-} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 2^+} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 2} f(x) = \underline{\hspace{2cm}},$$

$$\lim_{x \rightarrow 3^-} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 3^+} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 3} f(x) = \underline{\hspace{2cm}},$$

$$\lim_{x \rightarrow 4^-} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 4^+} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 4} f(x) = \underline{\hspace{2cm}}.$$

Example 7: Investigate limits of

$$f(x) = \sin\left(\frac{1}{x}\right)$$

Example 8: Graphically investigate limits of $\frac{x^2-1}{x^2-x-2}$

Suggested after class problems from APEX: Section 1.1 problems 9,11,15

1.3 - Finding Limits Analytically

Goals: Evaluate limits using limit laws, algebra, and the Squeeze Theorem.

Before Class Videos: 1.3.1–1.3.3

Definition:

If $f(x)$ gets arbitrarily close to L for all x sufficiently close to a , then

$$\lim_{x \rightarrow a} f(x) = L.$$

For many “nice” functions, evaluating a limit just means plugging in the number. We’ll investigate what functions are “nice” in section 1.5 on Continuity. For example:

Example 1:

- $\lim_{x \rightarrow 2} x^3 =$
- $\lim_{x \rightarrow \pi} \sin(x) =$
- $\lim_{x \rightarrow e} \ln(x) =$
- $\lim_{x \rightarrow 4} \sqrt{x} =$

Theorem:

If L , M , c , and k are real numbers and

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M,$$

then

- | | |
|----------------------------|---|
| 1. Sum Rule: | $\lim_{x \rightarrow c} [f(x) + g(x)] = L + M$ |
| 2. Difference Rule: | $\lim_{x \rightarrow c} [f(x) - g(x)] = L - M$ |
| 3. Constant Multiple Rule: | $\lim_{x \rightarrow c} [kf(x)] = kL$ |
| 4. Product Rule: | $\lim_{x \rightarrow c} [f(x)g(x)] = LM$ |
| 5. Quotient Rule: | $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$ |
| 6. Power Rule: | $\lim_{x \rightarrow c} [f(x)]^n = L^n, \quad n \text{ a positive integer}$ |
| 7. Root Rule: | $\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, \quad n \text{ a positive integer}$ |

If n is even in the Root Rule, assume that $f(x) \geq 0$ on an interval containing c .

Example 2: Evaluate

$$\lim_{x \rightarrow 2} \frac{x^2 - 2x}{x + 2}.$$

Example 3: Come up with functions f and g such that

$$\lim_{x \rightarrow 3} f(x)g(x) \neq \left[\lim_{x \rightarrow 3} f(x) \right] \left[\lim_{x \rightarrow 3} g(x) \right].$$

Four Algebraic Tricks

Example 4:

$$\lim_{x \rightarrow 2} \frac{x^2 - 2x}{x - 2}.$$

Example 5:

$$\lim_{x \rightarrow 0} \frac{(x - 2)^2 - 4}{x}.$$

Example 6:

$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}.$$

Example 7:

$$\lim_{x \rightarrow 2} \frac{\frac{1}{2} - \frac{1}{x}}{x - 2}.$$

Example 8: Evaluate

$$\lim_{x \rightarrow 4} \frac{4 - x}{5 - \sqrt{x^2 + 9}}.$$

Theorem:

Squeeze Theorem. If $f(x) \leq g(x) \leq h(x)$ for all x in an interval containing a (except possibly at a), and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L,$$

then

$$\lim_{x \rightarrow a} g(x) = L.$$

Example 9: Evaluate

$$\lim_{x \rightarrow 0} x^2 \sin(1/x^2).$$

Example 10: Suppose

$$-x^2 + 4x + 1 \leq g(x) \leq -2x + 10$$

for all x . For what value of a can you determine $\lim_{x \rightarrow a} g(x)$? What is the limit?

1.5 - Continuity

Goals: Determine where a function is continuous, build continuous piecewise functions, and apply the Intermediate Value Theorem.

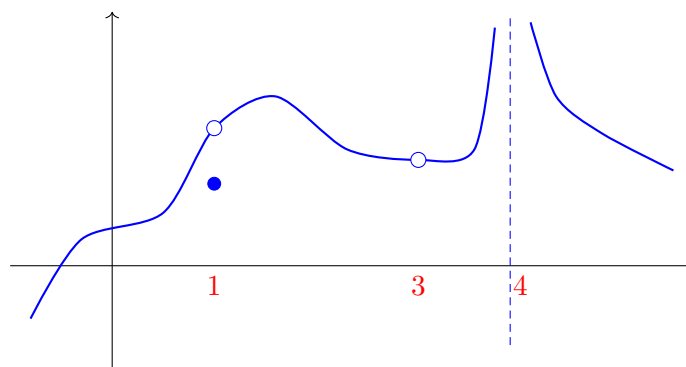
Before Class Videos: Section 1.5 videos 1 and 3

Definition:

A function f is **continuous at** c if all three of the following hold:

1. $f(c)$ exists,
2. $\lim_{x \rightarrow c} f(x)$ exists, and
3. $\lim_{x \rightarrow c} f(x) = f(c)$.

Example 1: Identify the type of discontinuity at each marked x -value.



Example 2: For what values of a is

$$f(x) = \begin{cases} x + 2, & x \leq 1, \\ ax^2, & x > 1 \end{cases}$$

continuous everywhere?

Example 3: For what values of a and b is

$$f(x) = \begin{cases} -2, & x \leq -1, \\ ax - b, & -1 < x < 1, \\ 3, & x \geq 1 \end{cases}$$

continuous everywhere?

Combining Continuous Functions

Theorem:

If f and g are continuous at c , then $f + g$, $f - g$, constant multiples of f , and fg are continuous at c . The quotient f/g is also continuous at c provided $g(c) \neq 0$.

Let's justify the sum rule from the limit laws:

Composition of Functions

Example 4: Compute the domain and range of $\sin \sqrt{x}$

Theorem:

Suppose f is continuous at a and g is continuous at $f(a)$. Then the composition $g \circ f$ is continuous at a , and

$$\lim_{x \rightarrow a} g(f(x)) = g(f(a)).$$

Example 5: On what intervals is $\sin\left(\frac{x-2}{x^2+5x+7}\right)$ continuous?

Example 6: Evaluate

$$\lim_{x \rightarrow 1} \frac{\sin(\pi x/6)e^{x^3} + x}{x^2 + 2}.$$

Theorem:

Intermediate Value Theorem. If f is continuous on $[a, b]$ and y_0 is between $f(a)$ and $f(b)$, then there is some $c \in [a, b]$ such that

$$f(c) = y_0.$$

Sketch:

Example 7: Sketch examples where the assumptions of the Intermediate Value Theorem are not satisfied.

Example 8: Show that

$$x^3 - 3x - 1 = 0$$

has a solution in $[0, 2]$.

Example 9: Use the intermediate value theorem to show that at some time a baby's height equalled their child's height.

Suggested after class problems from APEX: Section 1.5 problems 21,24,35

1.6 - Limits Involving Infinity

Goals: Investigate the special $\sin(x)/x$ limit, compute limits whose values are infinity, as well as limits at infinity

Before Class Videos: 1.6.1-1.6.3

Example 1: Explore $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

Example 2: Compute $\lim_{x \rightarrow 0} \frac{\sin(5x)}{\sin(3x)}$.

Example 3: Explore

$$f(x) = \frac{x}{x-2}$$

and find all asymptotes.

Definition:

$\lim_{x \rightarrow a} f(x) = \infty$ if $f(x)$ can be made arbitrarily large for all values of x sufficiently close to a .

Note: A limit being infinite is a particular example of a limit not existing.

Definition:

$\lim_{x \rightarrow \infty} f(x) = L$ if $f(x)$ can be made arbitrarily close to L for all values of x sufficiently large.

Example 4:

$$\lim_{x \rightarrow \infty} \sin(1/x)$$

Example 5:

$$\lim_{x \rightarrow \infty} x \sin(1/x)$$

Example 6: Find all Horizontal Asymptotes for

$$\frac{2x^3 + 3x}{x^4 - x^2}.$$

Example 7: Find all Horizontal Asymptotes for

$$\frac{\sqrt{x^6 + 2x}}{x^3 + x^{1/3}}.$$

Example 8: Find VA's; HA for $e^{1/x}$; graph it!

Example 9: Give example of a function with two HAs, $y = 0$ and $y = 2$, continuous everywhere.
Hint: $\arctan(x)$.

Suggested after class problems from APEX: Section 1.4 #40 and Section 1.6 #10, 19, 27

2.1 - Derivatives

Goals: Define the Derivative and apply it to initial examples

Before Class Videos: 2.1.1, 2.1.2

Example 1: Recall the velocity problem from the first day of class. If Trefor has some distance function $f(t)$, how fast is he going at 10 minutes? Let's explore graphically and numerically:

Definition:

The derivative of $f(x)$ at c is given by

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

provided this limit exists. If it does, we say f is differentiable at c .

Example 2: Compute the derivative of $f(x) = x^2$ at $x = 2$.

Definition:

The derivative of $f(x)$ on an open interval I is the function $f'(x)$ given by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists for all x in I . If it does we say $f(x)$ is differentiable on I .

Example 3: $f(x) = 1/x^2$.

Example 4: $f(x) = \sqrt{x}$.

Example 5: Let's explore if the following functions are differentiable at $x = 0$:

a)

$$f(x) = \begin{cases} 2, & x \leq 0, \\ 1, & x > 0. \end{cases}$$

b) $f(x) = |x|$

c) $f(x) = x^{2/3}$

d) $f(x) = x^{1/3}$

2.2 - Derivatives

Goals: A shorter than normal worksheet with a few extras on derivatives

Before Class Videos: Optional tricky example from last class's content: 2.1.4

Definition:

The derivative of $f(x)$ at c is given by

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

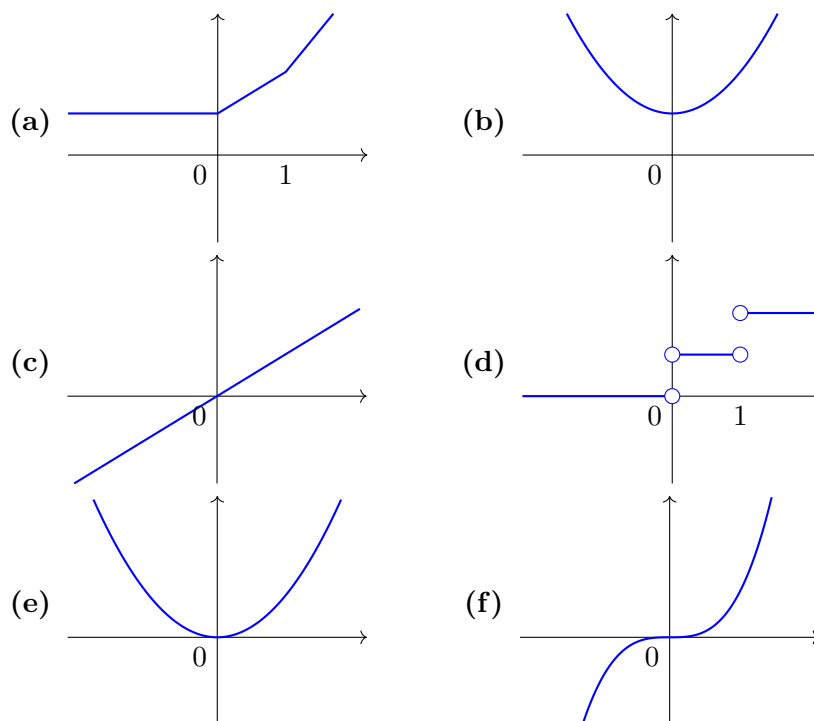
provided this limit exists. If it does, we say f is differentiable at c .

Definition:

An alternate formula for the derivative of $f(x)$ at c is

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}.$$

Example 1: The following six graphs consists of three pairs of functions and their derivative. Match them up.



Theorem:

Differentiable $\implies$ Continuous.

Note: This worksheet is deliberately short. I anticipate this class being a bit behind and finishing up prior material. If I am wrong I will edit it and add to it. The next worksheet is a more normal length.

Suggested after class problems from APEX: Section 2.2 15,17

2.3 - Differentiation Rules

Goals: State and apply Derivative Rules

Before Class Videos: 2.3.1

Theorem (Power Rule):

- $\frac{d}{dx}(c) = 0$ where c is a constant.
- $\frac{d}{dx}(x^n) = nx^{n-1}$ where $n \neq 0$

Theorem:

If $f(x)$ and $g(x)$ are differentiable at $x = a$, then

1. **Sum/Difference Rule:**

$$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$$

2. **Constant Multiple Rule:**

$$\frac{d}{dx}[cf(x)] = c\frac{d}{dx}f(x)$$

Example 1: Compute the derivative of $f(x) = 3x^4 - 5\sqrt{x} + 6x^{3/2} - \frac{7}{x^2} + 1$.

Proof of Theorem:

Definition:

The Point-Slope formula for the equation of the straight line through (x_0, y_0) with slope m is

$$y = y_0 + m(x - x_0)$$

Example 2: Find the equation of tangent line to a differentiable function $f(x)$ at the point x_0 .

Example 3: Find the equation of the tangent line to the function $f(x) = 2x^2 + 1$ at the point $(1, 3)$.

Common Derivatives:

Example 4: Explore the graphs of the following functions to discover their derivatives.

- $\frac{d}{dx}(e^x) =$

- $\frac{d}{dx} \ln(x) =$ for $x > 0$

- $\frac{d}{dx} \sin(x) =$

- $\frac{d}{dx} \cos(x) =$

Higher Derivatives: The derivative $f'(x)$ is itself a function we can take the derivative of. We call this the second derivative and might denote it by

$$f''(x) \quad \text{or} \quad \frac{d^2}{dx^2} f(x).$$

Example 5: Compute $f''(x)$ when $f(x) = e^x + \ln(x)$.

Example 6: If $f(x) = \sin(x)$ compute

- $f'(x) =$

- $f''(x) =$

- $f'''(x) =$

- $f^{(4)}(x) =$

- $f^{(2026)}(x) =$

Example 7: If $d(t)$ represents the position of a particle as a function of time, interpret $d'(t)$ and $d''(t)$.

Example 8: If $v(t)$ represents Trefor's total YouTube views after t days, interpret $v'(t)$ and $v''(t)$.

Suggested after class problems from APEX: Section 2.3 16, 21, 34 (tangent only)

2.4 - Product and Quotient Rule

Goals: Compute derivatives using the product and quotient Rules, including of trigonometric functions

Before Class Videos: 2.4.1 and 2.4.2

Theorem (Derivative Rules):

- $\frac{d}{dx}(c) = 0$ where c is a constant.
- $\frac{d}{dx}(x^n) = nx^{n-1}$ where $n > 0$ is an integer.

If $f(x)$ and $g(x)$ are differentiable at $x = a$, then

- $\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$
- $\frac{d}{dx}[cf(x)] = c\frac{d}{dx}f(x)$

Let's add two new rules to the list

Theorem (Product and Quotient Rules):

$$\begin{aligned}\frac{d}{dx}[fg] &= f'g + fg' \\ \frac{d}{dx}\left[\frac{f}{g}\right] &= \frac{f'g - fg'}{g^2}\end{aligned}$$

Example 1: Compute the following derivatives

a)

$$\sqrt{x}e^x$$

b)

$$\frac{\sin(x)}{x^3}$$

c)

$$\frac{xe^x}{x^2 + 1}$$

Example 2: Let's explore the intuition behind why the product rule is NOT just the product of the derivatives.

Example 3: Let's finally figure out what exactly e^x is all about. We'll use the definition of the derivative to compute, for $a > 0$,

$$\frac{d}{dx}a^x$$

Let's explore

$$\lim_{h \rightarrow 0} \frac{a^h - 1}{h}$$

Definition:

The number e is defined to be the value of a such that $\lim_{h \rightarrow 0} \frac{a^h - 1}{h}$

Right-Triangle Definitions

$$\begin{array}{lll} \sin x = \frac{\text{opposite}}{\text{hypotenuse}} & \tan x = \frac{\text{opposite}}{\text{adjacent}} & \sec x = \frac{\text{hypotenuse}}{\text{adjacent}} \\ \cos x = \frac{\text{adjacent}}{\text{hypotenuse}} & \cot x = \frac{\text{adjacent}}{\text{opposite}} & \csc x = \frac{\text{hypotenuse}}{\text{opposite}} \end{array}$$

Rewriting with Sine and Cosine

$$\begin{array}{lll} \sin x = \sin x & \tan x = \frac{\sin x}{\cos x} & \sec x = \frac{1}{\cos x} \\ \cos x = \cos x & \cot x = \frac{\cos x}{\sin x} & \csc x = \frac{1}{\sin x} \end{array}$$

Example 4: Let's explore the graphs of these functions.

Example 5: Compute $\frac{d}{dx} \tan(x)$

Example 6: Compute $\frac{d}{dx} \sec(x)$

Trigonometric Derivatives

$$\begin{array}{lll} \frac{d}{dx} \sin x = \cos x & \frac{d}{dx} \tan x = \sec^2 x & \frac{d}{dx} \sec x = \sec x \tan x \\ \frac{d}{dx} \cos x = -\sin x & \frac{d}{dx} \cot x = -\csc^2 x & \frac{d}{dx} \csc x = -\csc x \cot x \end{array}$$

Suggested after class problems from APEX: Section 2.4 7, 18, 26

2.5 - Chain Rule

Goals: Compute the derivative of compositions of functions

Before Class Videos: 2.5.1, 2.5.2

Example 1: Consider $\sin(x^2)$. This is an example of a composition $f(g(x))$ or alternatively $f \circ g$.

Example 2: $f(x) = x \sin x$, $g(x) = \cos x + 1$. Write out:

- $f(g(x)) =$

- $g(f(x)) =$

- $f(f(x)) =$

- $g(g(x)) =$

Theorem (Chain Rule):

If g differentiable at x and f differentiable at $g(x)$ then

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x).$$

Example 3: $\frac{d}{dx} \sin(x^2 + 1)$

Example 4: $\frac{d}{dx} \sin^{30}(x)$

Example 5: $\frac{d}{dx} \cos(3e^x)$

Example 6: $\frac{d}{dx} \frac{1}{2x+1}$

Example 7: $\frac{d}{dx} \cos^2(x^3)$

Example 8: $\frac{d}{dx} \tan(x^2 e^x)$.

Leibniz Notation: let $u = g(x)$,

$$\frac{d}{dx}[f(g(x))] = \frac{df}{du} \frac{du}{dx}.$$

Example 9: Car uses $6L/100km$. Cost is $1.5\$/L$. What is the rate of change in cost per $100km$?

Example 10: Gravel is dumped from a conveyor to form pile with

$$V(h) = \frac{\pi}{12}h^3.$$

At $5m$ tall, the pile is growing by 0.1 m/s. How fast is volume growing?

Example 11: Contrast graphically the slope of $f(x)$ and $f(2x)$ for an arbitrary function $f(x)$.

Example 12: Derive the Quotient Rule.

Suggested after class problems from APEX: Section 2.5: 7,17,24

2.6 - Implicit Differentiation

Goals: Compute derivatives of functions define implicitly

Before Class Videos: 2.6.1

Example 1: Explore $x^2 + y^2 = 1$. Where does it have vertical and horizontal tangents?

Big Idea: Treat $y = y(x)$ when x is independent and y is dependent.

Example 2: Find $\frac{dy}{dx}$ implicitly for $\sin(xy) + x = y^2 \cos x$.

Example 3: Compute $\frac{d^2y}{dx^2}$ for $x^2 + y^2 = 4$.

Example 4: Now explore $x^3 + y^3 - 9xy = 0$. Find all vertical and horizontal tangents.

Logarithmic Differentiation: Recall we previously learned that $\frac{d}{dx} \ln(x) = \frac{1}{x}$.

Example 5: $y = x^x$.

Example 6:

$$y = \frac{(x+2)^3(x-1)^{1/3}}{(x+7)^2}.$$

Example 7:

$$y = x^{\cos x} \cdot (\cos x)^x.$$